PRACTICE SHEET

1. Consider the following statements:
(i)The function f (x) = greatest integer ≤ x, x ∈ R is a continuous function.
(ii)All trigonometric functions are continuous on R.
Which of the statements given above is / are correct?
(a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) Neither 1 nor 2

2. If $f(x)=\begin{cases} mx+1 & x\le\frac{\pi}{2} \\ \sin x+n & x>\frac{\pi}{2}\end{cases}$ is continuous at $x\le\frac{\pi}{2}$, then which one of the following is correct?
(a) m = 1, n = 0 (b) $m=\frac{n\pi}{2}+1$
(c) $n=m\left(\frac{\pi}{2}\right)$ (d) $m=n=\frac{\pi}{2}$

3. If f (x) = $(x + 1)^{\cot x}$ is continuous at x = 0, then what is f (0) equal to?
(a) 1 (b) e
(c) 1/e (d) e^2

4. If the derivative of the function
$f(x)=\begin{cases} ax^2+b & x<-1 \\ bx^2+ax+4 & x\ge-1\end{cases}$
is everywhere continuous, then what are the values of a and b?
(a) a = 2, b = 3 (b) a = 3, b = 2
(c) a = –2, b = –3 (d) a = –3, b = –2

5. If f(x) is differentiable everywhere, then which one of the following is correct?
(a) |f| is differentiable everywhere
(b) $|f|^2$ is differentiable everywhere
(c) f|f| is not differentiable at some points
(d) None of the above

6. A function f is defined as follows
$f(x)=x^p\cos\left(\frac{1}{x}\right), x\ne0$ f(0) = 0
What conditions should be imposed on p so that f may be continuous at x = 0?
(a) p = 0 (b) p > 0
(c) p < 0 (d) No value of p

7. Let $f(x)=\begin{cases} 3x-4, & 0\le x\le2 \\ 2x+l, & 2<x\le9\end{cases}$
If f is continuous at x = 2, then what is the value of l ?
(a) 0 (b) 2
(c) – 2 (d) –1

8. Let f : R → be defined as f(x) = sin (|x|)
Which one of the following is correct?
(a) f is not differentiable only at 0
(b) f is differentiable at 0 only
(c) f is differentiable everywhere
(d) f is non – differentiable at many points

9. Consider the following function f:R→such that f(x) = x, if x ≥ 0 and f(x) = $-x^2$, if x < 0 then, which one of the following is correct?
(a)f(x) is continuous at every x∈R
(b)f(x) is continuous at x = 0 only
(c)f(x) is discontinuous at x = 0 only
(d)f(x) is discontinuous at every x∈R

10. The set of points where the function f(x)=x|x| is differentiable is:
(a) (–∞, ∞) (b) (–∞, 0)∪(0,∞)
(c) (0, ∞) (d) [0, ∞]

11. Consider the following statements:
I. $\lim_{x\to0}\frac{x^2}{x}$ exists.
II. $\left(\frac{x^2}{x}\right)$ is not continuous at x = 0.
III. $\lim_{x\to0}\frac{|x|}{x}$ does not exist
Which of the above statements are correct?
(a) I,II and III (b) I and II
(c)II and III (d) I and III

12. **Assertion (A):** $(x)=\begin{cases} x^2\sin\frac{1}{x}, & x\ne0 \\ 0, & x=0\end{cases}$ is continuous at x = 0
Reason (R): Both h(x) = x and $(x)=\begin{cases} x\sin\frac{1}{x}, & x\ne0 \\ 0, & x=0\end{cases}$ are continuous at x = 0.
(a) Both A and R are individually true, and R is the correct explanation of A.
(b) Both A and R are individually true but R is not the correct explanation of A.
(c) A is true but R is false
(d) A is false but R is true.

13. If the function f(x)=$\frac{x(x-2)}{x^2-4}$, x ≠ ±2 is continuous at x = 2, them what is f(2) equal to?
(a) 0 (b) $\frac{1}{2}$
(c) 1 (d) 2

14. At how many points is the function f(x) = [x] discontinuous?
(a) 1 (b) 2
(c) 3 (d) ∞

15. The function f(x) = x cosec x is
(a) Continuous at all value of x
(b) Discontinuous everywhere
(c) Continuous for all x except at x = nπ where n is an integer
(d) Continuous for all x except at x = nπ/2 where n is an integer

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OFFICIUM SUPREMUM

ANSWER KEYS																			
1.	d	**2.**	c	**3.**	b	**4.**	a	**5.**	c	**6.**	b	**7.**	c	**8.**	a	**9.**	a	**10.**	c
11.	d	**12.**	d	**13.**	b	**14.**	d	**15.**	c										

Solutions

Sol. 1. (d)
Here,
greatest integer function [x] is discontinuous at its integral value of x, cot x and cosec x are discontinuous at 0, π, 2π etc. and tan x and sec x are discontinuous at $x=\frac{\pi}{2},\frac{3\pi}{2},\frac{5\pi}{2}$ etc.
Therefore the greatest integer function and all trigonometric functions are not continuous for $x \in R$
Therefore, neither (1) nor (2) are true.

Sol. 2. (c)
Given function is

$$f(x)=\begin{cases} mx+1 & x\le \frac{\pi}{2} \\ \sin x+n, & x>\frac{\pi}{2}\end{cases}$$

As given this functions is continuous at $x\le\frac{\pi}{2}$

So, limit of function when $x\to\frac{\pi}{2}=f\left(\frac{\pi}{2}\right)$

$$\Rightarrow \lim_{x\to\frac{\pi}{2}}(\sin x+n)=f\left(\frac{\pi}{2}\right)$$

$$\Rightarrow \lim_{x\to 0}\left(\sin\left(\frac{\pi}{2}+h\right)+n\right)=\frac{m\pi}{2}+1$$

$$\Rightarrow \sin\frac{\pi}{2}+n=\frac{m\pi}{2}+1$$

$$\Rightarrow 1+n=\frac{m\pi}{2}+1$$

$$\Rightarrow n=\frac{m\pi}{2}$$

Sol. 3. (b)
For a function to be continuous at a point the limit should exist and should be equal to the value of the function at that point.
Here point is x = 0

and $\lim_{x\to 0}f(x)=\lim_{x\to 0}(x+1)^{\cot x}$

$$=\lim_{x\to 0}(1+x)^{\cot x}=\lim_{x\to 0}(1+x)^{\frac{1}{x}\cot x}$$

$$=\lim_{x\to 0}(1+x)^{\frac{1}{x}\lim_{x\to 0}\frac{x}{\tan x}}=e^1=e$$

Since limiting value of f(x) = e, when $x\to 0$, f(0) should also be equal to e.

Sol. 4. (a)
Derivative of $f(x)=\begin{cases} ax^2+b & x<-1 \\ bx^2+ax+a & x\ge -1\end{cases}$ is

$$f'(x)=\begin{cases} 2ax & x<-1 \\ 2bx+a & a\ge -1\end{cases}$$

If f'(x) is continuous everywhere then it is also continuous at x = –1

$$f'(x)\big|_{x=-1}=-2a=-2b+a$$

or, 3a = 2b
From the given choice
a = 2, b = 3 satisfied this equation.

Sol. 5. (c)
If f(x) is differential everywhere then |f| is not differentiable at some point, so, f |f| is not differentiable at some point.
[Example : f(x) = x is differentiable everywhere but |f(x)| = |x| is not differentiable at x = 0]

Sol. 6. (b)
Given function is defined as:

$$f(x)=\begin{cases} x^p\cos\left(\frac{1}{x}\right) & x\ne 0 \\ 0, & x=0\end{cases}$$

for continuity
LHS : $\lim_{x\to 0}f(x)=$ RHS $\lim_{x\to 0}f(x)=f(0)$

$$\Rightarrow \lim_{x\to 0}f(x)=\lim_{x\to 0}x^p\cos\left(\frac{1}{x}\right)=0$$

$$\Rightarrow \lim_{x\to 0}x^p\cos\left(\frac{1}{x}\right)=0$$

$\cos\left(\frac{1}{x}\right)$ is always a finite quantity if $x\to 0$

$\Rightarrow x^p=0$
Where is possible only if p > 0.

Sol. 7. (c)
Given function is

$$f(x)=\begin{cases} 3x-4 & 0\le x\le 2 \\ 2x+l & 2<x\le 9\end{cases}$$

and also given that f(x) is continuous at x = 2
For a function to be continuous at a point LHL = RHL
= V. F. at the point. F (2) = 2 = V.F.
$\Rightarrow$RHL : $\lim_{x\to 2}(2x+\ell)=3(2)-4$

$$\Rightarrow \lim_{x\to 0}\{2(2+h)+\ell\}=6-4$$

$\Rightarrow 4+l=2$
$\Rightarrow l=-2$

Sol. 8. (a)
Given functions is : f (x) = sin |x|

$$=\begin{cases}\sin(x) & x\ge 0 \\ \sin(-x), & x<0\end{cases}=\begin{cases}\sin x & x\ge 0 \\ -\sin x, & x<0\end{cases}$$

LHD at x = 0 = $\lim_{h\to 0}\frac{f(0-h)-f(0)}{0-h-0}$

$$=\lim_{h\to 0}\frac{f(0-h)-f(0)}{-h}=\lim_{h\to 0}\frac{-\sin(-h)-0}{-h}=-1$$

RHD at x = 0 $\lim_{h\to 0}\frac{f(0+h)-f(0)}{0+h-0}$

$$=\lim_{h\to 0}\frac{f(0+h)-f(0)}{h}=\lim_{h\to 0}\frac{-\sin(h-0)}{h}=1$$

LHD $\ne$ RHD
f(x) is not differentiable at x = 0

Sol. 9. (a)
Given, $f(x)=\begin{cases} x, & x\ge 0 \\ -x^2 & x<0\end{cases}$

LHL= $\lim_{x\to 0^-}f(x)=-\lim_{x\to 0}x^2=0$

RHL= $\lim_{x\to 0^+}f(x)=\lim_{x\to 0}x=0$

and $f(0)=0$
Since, LHL = RHL = $f(0)$
$\therefore f(x)$ is continuous at $x=0$
Also, $f(x)$ is continuous in the given interval ie. R
Hence, $f(x)$ is continuous in every $x\in R$

Sol. 10. (c)
Given, $f(x)=x\,|x|$
If $f(x_1)=f(x_2)$
$\Rightarrow x_1\,|x_1|=x_2\,|x_2|$
$\Rightarrow x_1=x_2$
$\therefore f(x)$ is one – one.
Also, range of $f(x)$ = co – domain of $f(x)$
$\therefore f(x)$ is onto
Hence, $f(x)$ is both one – one and onto

Sol. 11. (d)
1. $\lim_{x\to 0}\frac{x^2}{x^2}=x$ 2. $\frac{x^2}{x}=x$

Since, a polynomial is continuous everywhere, so it is continuous at $x=0$

3.LHL= $\lim_{h\to 0}\frac{|0-h|}{(0-h)}=\lim_{h\to 0}\frac{h}{-h}=-1$

RHL= $\lim_{h\to 0}\frac{|0+h|}{(0+h)}=\lim_{h\to 0}\frac{h}{h}=1$

$\because$ LHL $\ne$ RHL
So, it does not exist.
Thus, the statements 1 and 3 are correct.

Sol. 12. (d)
(A) : f(x) = log x for x = 1, f (x) = log 1 = 0
(R) : f (x) = log x
And f (x) $\ge$ 0 $\forall$ x > 0
Thus, (A) is false but (R) is true.

Sol. 13. (b)
Since, f is continuous at x = 2, therefore we have
$\lim_{x\to 2}f(x)$=f(2)

$$\Rightarrow \lim_{x\to 2}\frac{x(x-2)}{x^2-4}=f(2)$$

$$\Rightarrow \lim_{x\to 2}\frac{x}{x+2}=f(2)\Rightarrow(2)=\frac{2}{4}=\frac{1}{2}$$

Sol. 14. (d)
Sol. 15. (c)

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NDA PYQ

1. What is the value of k for which the following function f(x) is continuous for all x?

$$f(x)=\begin{cases}\dfrac{x^3-3x+2}{(x-1)^2}, & \text{for } x \neq 1\\ k, & \text{for } x = 1\end{cases}$$

(a) 3 (b) 2
(c) 1 (d) –1

[NDA (I) - 2011]

2. Which one of the following is correct in respect of the function $f(x)=|x|+x^2$?
(a)f(x) is not continuous at x = 0
(b)f(x) is differentiable at x = 0
(c)f(x) is continuous but not differentiable at x = 0
(d)None of the above

[NDA (I) - 2011]

3. Consider the following in respect of the function f(x) = |x–3|.
I.f(x) is continuous at x = 3
II.f(x) is differentiable at x = 0
Which of the above statement us/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA (I) - 2012]

4. Which one of the following functions is differentiable for all real values of x?
(a) $\dfrac{x}{|x|}$ (b) x|x|
(c) $\dfrac{1}{|x|}$ (d) $\dfrac{1}{x}$

[NDA (I) - 2012]

5. Which one of the following is correct in respect of the function $f(x)=\dfrac{x^2}{|x|}$ for x≠0 and f(0) = 0?
(a) f(x) is discontinuous everywhere
(b) f(x) is continuous everywhere
(c) f(x) is continuous at x = 0 only
(d) f(x) is discontinuous at x = 0 only

[NDA (II) - 2012]

6. Consider the following statements in respect of a function f(x)
I.f(x) continuous at x = a, if $\lim\limits_{x\to a} f(x)$ exists
II. If f(x) is continuous at a point, then $\dfrac{1}{f(x)}$ is also continuous at the point.
Which of the above statements is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA (I) - 2013]

7. The derivative of |x| at x = 0
(a) is 1 (b) is - 1
(c) is 0 (d) does not exist

[NDA (I) - 2013]

8. Consider the function

$$f(x)=\begin{cases}x^2 & \text{for } x > 2\\ 3x-2 & \text{for } x \leq 2\end{cases}$$

Which one of the following statements is correct in respect of the following function?
(a) f(x) is derivable but not continuous at x = 2
(b) f(x) is continuous but not derivable at x = 2
(c) f(x) is neither continuous nor derivable at x = 2
(d) f(x) is continuous as well as derivable at x = 2

[NDA (I) - 2013]

9. Consider the following statement
1. $f(x) = e^x$, where x > 0
2. g(x) = |x - 3|
Which of the above function is /are continuous?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) - 2013]

10. A function f:R→R is defined as $f(x) = x^2$ for x ≥ 0 and f(x) = -x for x < 0 consider the following statements in respect to the above function
1.The function is continuous at x = 0.
2.The function is differentiable at x = 0
Which of the above function is /are continuous?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) - 2013]

11. The following statements:
I.The function f(x) = $\sqrt[3]{x}$ is continuous at all x except at x = 0
II.The function f(x) = [x] is continuous at x = 2.99, where [.] is the bracket function.
Which of the above statements is/are correct?
(a) Only I (b) Only II
(c) Both I nor II (d) Neither I nor II

[NDA (I) - 2014]

12. Consider the following statements:
I.The function f(x)=|x| is not differentiable at x=1
II.The function $f(x) = e^x$ is differentiable at x = 0
Which of the above statements is/are correct?
(a) Only I (b) Only II
(c) Both I nor II (d) Neither I nor II

[NDA (I) - 2014]

Directions (for next three): Let f(x) be a function defined in 1 ≤ x < ∞ by.

$$f(x)=\begin{cases}2-x, & \text{for } 1 \leq x \leq 2\\ 3x-x^2, & \text{for } x > 2\end{cases}$$

13. Consider the function statements:
I.The function is continuous every point in the interval [1, ∞)
II.The function is differentiable at x = 15.
Which of the above statements is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA (I) - 2014]

14. What is the differentiable coefficient of f(x) at x = 3?
(a) 1 (b) 2
(c) –1 (d) –3

[NDA (I) - 2014]

15. Consider the following statements:

I.f'(2+0) does not exist
II.f'(2-0) does not exist
Which of the above statements is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
[NDA (I) - 2014]

Direction (for next two): Consider the function f(x)= $\frac{1-\sin x}{(\pi-2x)^2}$, where $x \neq \frac{\pi}{2}$ and $f\left(\frac{\pi}{2}\right)=\lambda$

16. What is $\lim_{x\to\frac{\pi}{2}} f(x)$ equal to?
(a) 1 (b) 1/2
(c) 1/4 (d) 1/8
[NDA (I) - 2014]

17. What is the value of λ, if the function is continuous at $x=\frac{\pi}{2}$?
(a) 1/8 (b) 1/4
(c) 1/2 (d) 1
[NDA (I) - 2014]

Directions (for next three): Read the following information carefully and answer the questions given below:

Consider the function $f(x)=\begin{cases} x^2-5, & x\le 3 \\ \sqrt{x+13}, & x>3 \end{cases}$

18. What is $\lim_{x\to 3} f(x)$ equal to:
(a) 2 (b) 4
(c) 5 (d) 13
[NDA (II) - 2014]

19. Consider the following statements:
I.The function is discontinuous at x = 3
II.The function is not differentiable at x = 0
Which of the above statements(s) is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
[NDA (II) - 2014]

20. What is the differential coefficient of f(x) at x = 12?
(a) 5/2 (b) 5
(c) 1/5 (d) 1/10
[NDA (II) - 2014]

21. Consider the function: $f(x)=\begin{cases} \frac{\tan kx}{x}, & x<0 \\ 3x+2k^2 & x\le 0 \end{cases}$
What is the non-zero value of k for which the function is continuous at x = 0?
(a) 1/4 (b)1/2
(c) 1 (d) 2
[NDA (II) - 2014]

22. Consider the following statements:
I.The function f(x) = [x], where [.] is the greatest integer function defined on R, is continuous at all points except at x = 0.
II.The function f(x) = sin|x| is continuous for all x∈R.
Which of the above statements is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
[NDA (II) - 2014]

Directions (for next two): Consider the function:

$$f(x)=\begin{cases} \frac{\alpha\cos x}{\pi-2x}, & \text{if } x\neq\frac{\pi}{2} \\ 3, & \text{if } x=\frac{\pi}{2} \end{cases}$$

Which is continuous at x=π/2, where α is a constant.

23. What is the value of α?
(a) 6 (b) 3
(c) 2 (d) 1
[NDA (I) - 2015]

24. What is $\lim_{x\to 0} f(x)$ equal to?
(a) 0 (b) 3
(c) $\frac{3}{\pi}$ (d) $\frac{6}{\pi}$
[NDA (I) - 2015]

Directions (for next two): Given a function

$$f(x)\begin{cases} -1 & \text{if} \quad x\le 0 \\ ax+b, & \text{if} \quad 0<x<1 \\ 1, & \text{if} \quad x\ge 1 \end{cases}$$

Where a, b are constants. The function is continuous everywhere.

25. What is the value of b?
(a) –1 (b) 1
(c) 0 (d) 2
[NDA (I) - 2015]

26. What is the value of a?
(a) –1 (b) 0
(c) 1 (d) 2
[NDA (I) - 2015]

27. Consider the following statements:
I.f(x) = [x], where [.] is the greatest integer function, is discontinuous at x = n, where n∈Z.
II. f(x) cot x is discontinuous at x = nπ, where n∈Z.
Which of the above statements is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
[NDA (I) - 2015]

28. Consider the function

$$f(x)=\begin{cases} ax-2 & \text{for} \quad -2<x<-1 \\ -1 & \text{for} \quad -1\le x\le 1 \\ a+2(x-1)^2 & \text{for} \quad 1<x<2 \end{cases}$$

What is the value of a for which f(x) is continuous at x = –1 and x =1?
(a) –1 (b) 1
(c) 0 (d) 2
[NDA (II) - 2015]

29. The function $f(x)=\frac{1-\sin x+\cos x}{1+\sin x+\cos x}$ is not defined at x=π. The value of f(π) so that f(x) is continuous at x = π is:
(a) –1/2 (b)1/2
(c) –1 (d) 1
[NDA (II) - 2015]

30. Consider the following functions:

1.$f(x)=\begin{cases} \frac{1}{x} & \text{if } x\neq 0 \\ 0 & \text{if } x=0 \end{cases}$

2.$f(x)=\begin{cases} 2x+5 & \text{if } x>0 \\ x^2+2x+5 & \text{if } x\le 0 \end{cases}$

Which of the above function is/are derivable at x = 0?
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2
[NDA (II) - 2015]

Directions (for next two): Consider the function

$$f(x) = \begin{cases} -2\sin x & \text{for} \quad x \le -\frac{\pi}{2} \\ A\sin x + B & \text{for} \quad -\frac{\pi}{2} \le x \le \frac{\pi}{2} \\ \cos x & \text{for} \quad x \ge \frac{\pi}{2} \end{cases}$$

Which is continuous at everywhere

31. The value of A is
(a) 1 (b) 0
(c) -1 (d) -2
[NDA (II) - 2015]

32. The value of B is
(a) 1 (b) 0
(c) -1 (d) -2
[NDA (II) - 2015]

Direction (for next two): Consider the following for the next two items that follows

$$f(x) = \begin{cases} x + \pi \text{ for } x \in [-\pi, 0) \\ \pi \cos x \text{ for } x \in \left[0, \frac{\pi}{2}\right] \\ \left(x - \frac{\pi}{2}\right)^2 \text{ for } x \in \left(\frac{\pi}{2}, \pi\right] \end{cases} \backslash$$

33. Consider the following statement:
1.The function f(x) is continuous at x = 0.
2.The function f(x) is continuous at x= $\frac{\pi}{2}$
Which of the above statements is/are correct?
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2
[NDA (I) - 2016]

34. Consider the following statements:
1.The function f(x) is differentiable at x = 0.
2.The function f(x) is differentiable at x = $\frac{\pi}{2}$
Which of the above statements is / are correct?
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2
[NDA (I) - 2016]

35. Consider the function
$f(x)=|x-1| + x^2$ where $x \in R$.
Which one of the following statements is correct?
(a) f(x) is continuous but not differentiable at x=0
(b) f(x) is continuous but not differentiable at x=1
(c) f(x) is differentiable at x=1
(d) f(x) is not differentiable at x=0 and x=1
[NDA-2016(1)]

Direction (for next three): Consider the following for the next two items that follow:
Let f(x) = [x], where [.] is the greatest integer function and g(x) = sin x be two real valued function over R.

36. Which of the following statements is correct?
(a) Both f(x) and g(x) are continuous at x = 0
(b) f(x) is continuous at x = 0, but g(x) is not continuous at x = 0
(c) g(x) is continuous at x = 0, but f(x) is not continuous at x = 0
(d) Both f(x) and g(x) are discontinuous at x =0
[NDA (II) - 2016]

37. Which one of the following statements is correct?
(a) $\lim_{x\to 0}$ (fog) (x) exists
(b) $\lim_{x\to 0}$ (gof) (x) exists
(c) $\lim_{x\to 0^-}$ (fog) (x) = $\lim_{x\to 0^-}$ (gof) (x)
(d) $\lim_{x\to 0^+}$ (fog) (x) = $\lim_{x\to 0^+}$ (gof) (x)
[NDA (II) - 2016]

38. Which of the following statements are correct?
1. (fof)(x) = f(x)
2. (gog)(x) = g(x) only when x = 0
3. (go(fog)(x) can take only three values
select the correct answer using the code given below
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d)1, 2 and 3
[NDA (II) – 2016]

39. Consider the following in respect of the function f(x) = $\begin{cases} 2+x, & x \ge 0 \\ 2-x, & x < 0 \end{cases}$
1. $\lim_{x\to 1}$ f(x) does not exist
2. f(x) is differentiable at x = 0
3. f(x) is continuous at x = 0
Which of the above statements is / are correct?
(a) Only 1 (b) Only 3
(c) Both 2 and 3 (d) Both 1 and 3
[NDA (II) - 2016]

Direction (for next three): Consider the following for the next two items that follow:
Let f(x) = $\begin{cases} -2 & -3 \le x \le 0 \\ x-2, & 0 < x \le 3 \end{cases}$ and
g(x) = f (|x|) + |f(x)|

40. Which of the following statements is/ are correct?
1.g(x) is differentiable at x = 0
2. g(x) is differentiable at x = 2
Select the correct answer using the codes given below:
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2
[NDA (II) - 2016]

41. What is the value of the differential coefficient of g(x) at x = –2?
(a) –1 (b) 0
(c) 1 (d) 2
[NDA (II) - 2016]

42. Which of the following statements are correct?
1. g(x) is continuous at x = 0
2. g(x) is continuous at x = 2.
3. g(x) is continuous at x = –1.
Select the correct answer using the code given below:
(a) Both 1 and 2 (b) Both 2 and 3
(c) Both 1 and 3 (d) 1, 2 and 3
[NDA (II) - 2016]

43. Let f : A → R, where A = R/(0) is such that f(x) = $\frac{x+|x|}{x}$.
On which one of the following sets is f(x) continuous?
(a) A (b) B = $\{x \in R : x \ge 0\}$

(c) $C = \{x \in R : x \le 0\}$ (d) $D = R$

[NDA (II) - 2016]

44. Suppose the function $f(x) = x^n$, $n \neq 0$ is differentiable for all x. then n can be any element of the interval
(a) $[1,\infty)$ (b) $(0,\infty)$
(c) $(1/2,\infty)$ (d) none of the above

[NDA (I) - 2017]

45. Let f (x) be defined as follows:

$$f(x) = \begin{cases} 2x+1 & -3 < x < -2 \\ x-1 & -2 \le x < 0 \\ x+2 & 0 \le x < 1 \end{cases}$$

Which one of the following statements is correct in respect of the above function?
(a) If is discontinuous at $x = -2$ but continuous at every other point.
(b) It is continuous only in the interval $(-3, -2)$
(c)It is discontinuous at $x = 0$ but continuous at every other point.
(d) It is discontinuous at every point.

[NDA (I) - 2017]

46. A function is defined as follows:

$$f(x) = \begin{cases} -\dfrac{x}{\sqrt{x^2}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Which one of the following is correct in respect of the above function?
(a)f(x) is continuous at x = 0 but not differentiable at x = 0
(b)f(x) is continuous as well as differentiable at x = 0
(c)f(x) is discontinuous at x = 0
(d)None of the above

NDA (II) - 2017]

47. If $f(x) = x\left(\sqrt{x} - \sqrt{x+1}\right)$, then f(x) is:
(a) Continuous but not differentiable at x = 0
(b) Differentiable at x = 0
(c) Not continuous at x = 0
(d) None of the above

[NDA (II) - 2017]

48. Consider the following :
1. $x + x^2$ is continuous at x = 0
2. $x + \cos 1/x$ is discontinuous at x = 0
3. $x^2 + \cos 1/x$ is continuous at x = 0
Which of the above are correct?
(a) 1 and 2 (b) 2 and 3
(c) 1 and 3 (d) 2, 2 and 3

[NDA (II) - 2017]

49. The left-hand derivative of $f(x) = [x] \sin(\pi x)$ at x=k
Where k is an integer and [x] is the greatest integer function, is:
(a) $(-1)^k (k-1)\pi$ (b) $(-1)^{k-1} (k-1)\pi$
(c) $(-1)^k k\pi$ (d) $(-1)^{k-1} k\pi$

[NDA-2017(2)]

50. If $f(x) = x/2-1$, then on the interval $[0, \pi]$ which one of the following is correct?
(a)tan [f(x)], where [.] is the greatest integer function, and 1/f(x) are both continuous.
(b)tan [f(x)], where [.] is the greatest integer function, and $f^{-1}(x)$ are both continuous
(c)tan [f(x)], where [.] is the greatest integer function, and 1/f(x) are both discontinuous
(d)tan [f(x)], where [.] is the greatest integer function, is discontinuous but 1/f(x) is continuous.

[NDA-2017(2)]

51. The set of all points, where the function $f(x) = \sqrt{1-e^{-x^2}}$ is differentiable, is:
(a) $(0, \infty)$ (b) $(-\infty, \infty)$
(c) $(-\infty, 0) \cup (0, \infty)$ (d) $(-1, \infty)$

[NDA-2017(2)]

52. Let g be the greatest integer function. Then the function $f(x)=(g(x))^2 - g(x^2)$ is discontinuous at:
(a) all integers
(b) all integers except 0 to 1
(c) all integers except 0
(d) all integers except 1

[NDA-2017(2)]

53. If $f(x)=\dfrac{x^2-9}{x^2-2x-3}$, $x \neq 3$ is continuous at x = 3, then which one of the following is correct?
(a) f(3)=0 (b) f(3) = 1.5
(c) f(3)=3 (d) f(3)= –1.5

[NDA (I) - 2018]

54. If $f(x) = |x| + |x-1|$, then which one of the following is correct?
(a)f(x) is continuous at x = 0 and x = 1
(b)f(x) is continuous at x = 0 but not at x = 1
(c)f(x) is continuous at x = 1 but not at x = 0
(d)f(x) is neither continuous at x = 0 nor at x=1

[NDA (I) - 2018]

55. If the function $f(x) = \dfrac{2x - \sin^{-1} x}{2x + \tan^{-1} x}$ is continuous at each point in its domain, then what is the value of f(0)?
(a) –1/3 (b) 1/3
(c) 2/3 (d) 2

[NDA (II) - 2018]

56. For the function $f(x) = |x-3|$, which of the following is not correct?
(a)The function is not continuous at x = –3
(b)The function is continuous at x = 3
(c)The function is differentiable at x = 0
(d)The function is differentiable at x= –3

[NDA (II) - 2018]

57. Consider the function:

$$f(x) = \begin{cases} \dfrac{\sin 2x}{5x} & \text{if } x \neq 0 \\ \dfrac{2}{15} & \text{if } x = 0 \end{cases} :$$

Which one of the following is correct in respect of the function?
(a) It is not continuous at x = 0
(b) It is continuous at every x
(c) It is not continuous at x= π
(d) It is continuous at x = 0

[NDA (II) - 2018]

58. If the function $f(x) = \begin{cases} \sin x, & x \neq 0 \\ k & x = 0 \end{cases}$ is continuous at x = 0 than find the value of k?
(a) 2 (b) 1
(c) - 1 (d) 0

[NDA (I) - 2019]

59. For what value of k is the function

$$f(x) = \begin{cases} 2x + \frac{1}{4} & ,x < 0 \\ k & ,x = 0 \\ \left(x + \frac{1}{2}\right)^2 & ,x > 0 \end{cases}$$ continuous?

(a) 1/4 (b) 1/2
(c) 1 (d) 2

[NDA (II) - 2019]

60. Consider the following statements in respect of the function $f(x) = \sin\left(\frac{1}{x}\right)$ for $x \neq 0$ and $f(0)=0$
1. $\lim_{x\to 0} f(x)$ exists
2. f(x) is continuous at x = 0

Which of the above statements is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) - 2019]

61. Consider the following statements for $f(x) = e^{-|x|}$
1. The function is continuous at x = 0.
2. The function is differentiable at x =0

Which of the above statements is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA 2020]

62. $f(x) = \frac{\sin x}{x}$, where $x \in R$, is to be continuous at x = 0, then the value of the function at x = 0
(a) should be 0
(b) should be 1
(c) should be 2
(d) cannot be determined

[NDA 2020]

63. If the function
$$f(x) = \begin{cases} a + bx, & x < 1 \\ 5, & x = 1 \\ b - ax, & x > 1 \end{cases}$$
is continuous, then what is the value of (a + b)?
(a) 5 (b) 10
(c) 15 (d) 20

[NDA (I) - 2021]

64. Consider the following statements in respect of $f(x) = |x| - 1$:
1. f(x)is continuous at x = 1
2. f(x)is differentiable at x = 0

Which of the above statements is/are correct?
(a)1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 and 2

[NDA (II) - 2021]

65. Consider the following statements in respect of the function $f(x)=\sin\left(\frac{1}{x^2}\right)$, $x \neq 0$:
1.It is continuous at x = 0, if f(0) = 0
2.It is continuous at $x=\frac{2}{\sqrt{\pi}}$

Which of the above statements is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) - 2021]

66. Consider the following statements in respect of the function $y = [x]$, $x \in (-1, 1)$ where [.] is the greatest integer function:
1. Its derivative is 0 at x = 0.5
2.It is continuous at x = 0

Which of the above statement is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) - 2022]

Consider the following for the next two (02) items that follow:

Let $f(x) = \begin{cases} x + 6, x \le 1 \\ px + q, 1 < x < 2 \\ 5x, \ x \ge 2 \end{cases}$ and f(x) is continuous.

67. What is the value of p?
(a) 2 (b) 3
(c) 4 (d) 5

[NDA – 2023 (1)]

68. What is the value of q?
(a) 2 (b) 3
(c) 4 (d) 5

[NDA – 2023 (1)]

Consider the following for the next (02) items that follow

Let $f(x) = \begin{cases} \frac{x-3}{|x-3|} + a; & x < 3 \\ a - b; & x = 3 \text{ and} \\ \frac{x-3}{|x-3|} + b; & x > 3 \end{cases}$

f(x) be continuous at x = 3.

69. What is the value of a?
(a) –1 (b) 1
(c) 2 (d) 3

[NDA-2023 (2)]

70. What is the value of b?
(a) –1 (b) 1
(c) 2 (d) 3

[NDA-2023 (2)]

Consider the following for the next (02) items that follow

Let $f(x)=\begin{cases} ax(x+1)+b & x<1 \\ x-1, & 1 \le x \le 2 \end{cases}$

71. If the function f(x) is differentiable at x=1, then what is the value of (a+b)?
(a) –1/3 (b) –1
(c) 0 (d) 1

[NDA-2023 (2)]

72. What is $\lim_{x\to 0} f(x)$ equal to?
(a) –1/3 (b) –2/3
(c) 0 (d) 1

[NDA-2023 (2)]

73. Let $f(x) = |x| + 1$ and $g(x) = [x] - 1$,. Where [.] is the greatest integer function

Let $h(x) = \frac{f(x)}{g(x)}$

Consider the following statements:
1. f(x) is differentiable for all x < 0
2. g(x) is continuous at x = 0.0001
3. The derivative of g(x) at x = 2.5 is 1

Which of the statements given above are correct?
(a) 1 and 2 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

[NDA-2024 (1)]

Direction: Consider the following for the **one items** given below

Let $f(x) = [x]^2 - [x^2]$

74. Consider the following statements:

I. f(x) is continuous at x = 0

II. f(x) is continuous at x = 1

Which of the statements given above is/are correct?

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

[NDA-2024 (2)]

Consider the following for the two (02) items that follow:

Let $f(x) = \begin{cases} x^3 & x^2 < 1 \\ x^2, & x^2 \geq 1 \end{cases}$

75. What is $\lim_{x \to 0} f'(x)$ equal to?

(a) 2 (b) 1

(c) 0 (d) Limit does not exist

[NDA-2025 (1)]

76. Consider the following statements:

I. The function is continuous at x = − 1.

II. The function is differentiable at x = 1.

Which of the statements given above is/are correct?

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

[NDA-2025 (1)]

For the following next **one** items:

Consider the function

$$f(x) = \begin{cases} 4(5^x) & x < 0 \\ 8k + x & x \geq 0 \end{cases}$$

77. If the function is continuous, then what is the value of k?

(a) 0.5 (b) 1

(c) 1.5 (d) 2

[NDA-2025 (2)]

For the following **one** items:

Let the function f(x) = |x − 3| + |x − 4| be defined on the interval [0,5].

78. Consider the following statements:

I. The function is differentiable at x = 3

II. The function is differentiable at x = 4

Which of the statements given above is/are correct?

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

[NDA-2025 (2)]

ANSWER KEY

1.	a	**2.**	c	**3.**	c	**4.**	b	**5.**	b	**6.**	d	**7.**	d	**8.**	b	**9.**	c	**10.**	a
11.	b	**12.**	b	**13.**	b	**14.**	d	**15.**	d	**16.**	d	**17.**	a	**18.**	b	**19.**	d	**20.**	d
21.	b	**22**	b	**23.**	a	**24.**	d	**25.**	a	**26.**	d	**27.**	c	**28.**	a	**29.**	c	**30.**	b
31.	c	**32.**	a	**33.**	c	**34.**	d	**35.**	b	**36.**	c	**37.**	d	**38.**	d	**39.**	b	**40.**	d
41.	a	**42.**	d	**43.**	a	**44.**	a	**45.**	c	**46.**	c	**47.**	c	**48.**	a	**49.**	a	**50.**	c
51.	c	**52.**	d	**53.**	b	**54.**	a	**55.**	b	**56.**	a	**57.**	a	**58.**	d	**59.**	a	**60.**	d
61.	a	**62.**	b	**63.**	a	**64.**	a	**65.**	b	**66.**	a	**67.**	b	**68.**	c	**69.**	d	**70.**	b
71.	c	**72.**	c	**73.**	a	**74.**	b	**75.**	c	**76.**	d	**77.**	a	**78.**	d				

OFFICIUM SUPREMUM

Solutions

Sol. 1. (a)
Continuous for all x.
$\therefore \lim_{x\to1} f(x) = f(1) \Rightarrow \lim_{x\to1} \frac{x^3-3x+2}{(x-1)^2} = k \left[\frac{0}{0}\text{form}\right]$
By L' Hospital rule, we get, k =
$\lim_{x\to1} \frac{3x^2-3}{2(x-1)} \left[\frac{0}{0}\text{form}\right]$
Again, by L' Hospital rule, we get
k= $\lim_{x\to1} \frac{6x}{2} = 3.(1) \Rightarrow k=3$

Sol. 2. (c)
Some of two continuous functions is always continuous and sum of differentiable and not differential able is not differential.

Sol. 3. (c)
Given f (x) = |x – 3|
Modulus functions are always continuous but not differentiable at point where f(x) is 0.
Statement first is correct.
Statement Second is also correct.

Sol. 4. (b)
Clearly, the function f(x) = x |x| is differentiable, as all other functions are not defined for x = 0.

Sol. 5. (b)
We have
f(x)= $\frac{x^2}{|x|}$, x≠0 and f(0) = 0
or fx = $\begin{cases} \frac{x^2}{x} = -x & \text{if } x > 0 \\ \frac{x^2}{-x} = -x & \text{if } x < 0 \\ 0, & \text{if } x = 0 \end{cases}$
Clearly, it is modulus functions and modulus function is continuous everywhere.

Sol. 6. (d)
I.We know that if f(a) = $\lim_{x\to a}$ f(x), then f(x) is continuous at x = a. but f(a) is not given in question.
II.If f(x) is continuous at a point, then it is not necessary that $\frac{1}{f(x)}$ is also continuous at the point.
For example
(i) f(x) = x is continuous at x = 0 but f (x) = 1/x is not continuous at x = 0
(ii)f(x) = e^x is continuous at x = 0 and f(x) = e^{-x} is also continuous at x = 0.

Sol. 7. (d)
Let f(x) = |x|⇒f(x)= $\begin{cases} x & x > 0 \\ 0, & x = 0 \\ -x & x < 0 \end{cases}$
Now, consider, LHD= Lf'(0)= $\lim_{h\to0} \frac{f(0-h)-f(0)}{-h}$
$= \lim_{h\to0} \frac{-(-h)-0}{-h} = \lim_{h\to0} \frac{h}{-h} = -1$
and RHD=Rf'(0) =
$\lim_{h\to0} \frac{f(0+h)-f(0)}{h} = \lim_{h\to0} \frac{h-0}{h} = 1 \because$
Lf'(0) ≠ Rf'(0)
∴ The derivative of |x| at x = 0 does not exist.

Sol. 8. (b)
Given a function
$f(x) = \begin{cases} x^2 & x > 2 \\ 3x-2 & x \le 2 \end{cases}$
First we check the continuity of f (x) at x = 2. For this Cosider
LHL=f(2–0)= $\lim_{h\to0}$ f(2–h)
= $\lim_{h\to0}$ [2(2–h)–2]= $\lim_{h\to0}$ (6–3h–2)
$\lim_{h\to0}$ (4–3h)=4–0=4
&RHL = f'(2+0)= $\lim_{h\to0}$ f(2+h)
= $\lim_{h\to0}$ $(2+h)^2 = (2+0)^2 = 4$
Also, f(2)=3(2)–2=6–2=4
Thus, we have f (2) = LHL = RHL
Hence, f(x) is continuous at x = 2
Now, we check the differentiability of f(x) at x = 2. For this Consider,
LHD=Lf'(2)= $\lim_{h\to0} \frac{f(2-h)-f(2)}{-h}$
$= \lim_{h\to0} \frac{3(2-h)-2-(3.2-2)}{-h}$
$= \lim_{h\to0} \frac{6-3h-2-6+2}{-h} = \lim_{h\to0} \frac{-3h}{-h} = 3$
and RHD=Rf'(2) = $\lim_{h\to0} \frac{f(2+h)-f(2)}{h}$
$= \lim_{h\to0} \frac{(2+h)^2-f(2)}{h} = \lim_{h\to0} \frac{4+h^2+4h-3(3.2.-2)}{h}$
$= \lim_{h\to0} \frac{4+h^2+4h-6+2}{h}$
= $\lim_{h\to0}$ (h+4)=0+4=4
∵ Lf' is not differentiable at x = 2
∴ f (x) is not differentiable at x = 2
Hence, f(x) is continuous but not derivable at x = 2

Sol. 9. (c)
I. f(x) = e^x where x > 0
II. function g (x) = |x – 3|

Sol. 10. (a)
Given a function
$f: R \to R, \text{defined as } f(x) = \begin{cases} x^2 & x \ge 0 \\ -x & x < 0 \end{cases}$
For continuity at x = 0, consider
LHL=f(0 – 0)= $\lim_{h\to0}$ f(0–h)
= $\lim_{h\to0}$ [- (0 – h)]= $\lim_{h\to0}$ f(0–h)
and RHL = f(0+0)= $\lim_{h\to0}$ f(0+h)= $\lim_{h\to0}$ $(0+h)^2$ = 0
also, f(0) = 0
∴f(0) = LHL = RHL Hence, f(x) is continuous at x = 0
For differentiability at x = 0, consider
L f'(0)= $\lim_{h\to0} \frac{f(0-h)-f(0)}{-h}$
$= \lim_{h\to0} \frac{-(-h)-0}{-h} = \lim_{h\to0} \frac{h}{-h} = -1$
And Rf'(0)= $\lim_{h\to0}$
$\frac{f(0+h)-f(0)}{h} = \lim_{h\to0} \frac{h^2-0}{h} = \lim_{h\to0} h = 0$
∵ L f' (0) ≠ R f' (0)
∴ f (x) is not differentiable at x = 0

Sol. 11. (b)
I. Given, $f(x) = \sqrt[3]{x}$
$\Rightarrow f(x) = (x)^{1/3}$
Now, we check the continuity of the function at x = 0.
LHL = f'(0–0)= $\lim_{h\to0} f(0-h) = \lim_{h\to0} (0-h)^{1/3}$
=$(0–0)^{1/3}$ = 0
RHL = f'(0+0)= $\lim_{h\to0} f(0-h) = \lim_{h\to0} (0-h)$
=$(0+0)^{1/3}$ = 0 and f(0) = $(0)^{1/3}$ =0
∵ So, function is continuous at x = 0
II. Given, f(x) = [x]
Which is the greatest integer function.
We know that, the greatest nteger function is continuous for all x except integer values of x.
So, f(x) = [x] is continuous at x = 2.99
Alternate Method
LHL = f'(2.99–0)= $\lim_{h\to0}$ f(2.99–h)
= $\lim_{h\to0}$ [2.99–h]= $\lim_{h\to0}$ 2=2
RHL=f(2.99+0)= $\lim_{h\to0}$ f(2.99+h)
= $\lim_{h\to0}$ [2.99+h]= $\lim_{h\to0}$ 2=2
and f(2.99)=[2.99]=2
∵ LHL = RHL = f(2.99)
So, f(x) is continuous at x = 2.99

Sol. 12. (b)
I. f (x) = |x|. we observe that, the curve has sharp turn at x = 0.
So, the function f (x) = |x| is not differentiable only at x = 0.
i.e., f (x) = |x| is differentiable at x = 1
II. Given function, f(x) = e^x
Now, we check the differentiability of f(x)at x = 0.
Consider, f'(0) = $\lim_{h\to0} \frac{f(0+h)-f(0)}{h}$
$= \lim_{h\to0} \frac{e^{(0+h)}-e^0}{h} = \lim_{h\to0} \frac{e^h-1}{h} \left[\frac{0}{0}\text{form}\right]$
$= \lim_{h\to0} \frac{e^h-0}{1} = e^0$ =1 [by L' Hospital's rule]
Hence, f(x) = e^x is differentiable at x = 0

Sol. 13. (b)

Since, the function is polynomial, so it is continuous as well as differentiable in the domain $[1, \infty) - \{2\}$

Now, we check the continuity of the function at x = 2

Consider, LHL= f' (2–0) = $\lim_{h\to 0} f(2-h)$

$= \lim_{h\to 0} 2-(2-h) = \lim_{h\to 0} h = 0$

and RHL = f(2+0)= $\lim_{h\to 0} f(2+h)$

$= \lim_{h\to 0} 3(2+h) - (2+h)^2$

$=3(2+0)-(2+0)^2 = 6 - 4 = 2$

Also, f(2) = 2–2 = 0

$\because$ f(2)= LHL ≠ RHL

So, the function is not continuous at every point in the interval $[1, \infty]$ as it is not continuous at x = 2.

According to above discussion we can say that f is differentiable at x = 1.5

Sol. 14. (d)

$$\because f(x) = \begin{cases} 2-x & \text{for } 1 \le x \le 2 \\ 3x-x^2 & \text{for } x > 2 \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} -1 & \text{for } 1 < x < 2 \\ 3-2x & \text{for } x > 2 \end{cases}$$

So, the differentiable coefficient of f(x) at = 3 is f' (3) = 3 – 2 (3) = 3 – 6 = –3

[$\because$ f' (x) = 3 – 2x, for x > 2]

Sol. 15. (d)

Here, we check the existence of limit of f' (x) at x = 2.

So, f'(2+0) = RHL= $\lim_{h\to 0} f'(2+h)$

$= \lim_{h\to 0} 3-2(2+h)$

$= \lim_{h\to 0} 3-4-2h = \lim_{h\to 0} -1-2h = -1$

and f'(2–0)=LHL= $\lim_{h\to 0} f'(2-h) = \lim_{h\to 0} -1 = -1$

since both f' (2+0) and f'(2–0) exist and equal, therefore f'(x) exist at x = 2

since both f' (2 + 0) and f' (2 – 0) exists and equal, therefore f'(x) exist at x = 2

Sol. 16. (d)

Consider,

$$\lim_{h\to\pi/2} f(x) = \lim_{h\to\pi/2} \frac{1-\sin(x)}{(\pi-2x)^2} \quad \left[\frac{0}{0}\text{form}\right]$$

$$= \lim_{h\to\pi/2} \frac{-\cos x}{(\pi-2x)(-2)} \text{ (by L' hospital' rule)}$$

$$= \lim_{h\to\pi/2} \frac{-\sin x}{4(\pi-2x)} \quad \left[\frac{0}{0}\text{form}\right]$$

$$= \lim_{h\to\pi/2} \frac{-\sin x}{4(-2)} = \lim_{h\to\pi/2} \frac{\sin x}{8} \text{ [by L' Hospital's rule]}$$

$$= \frac{1}{8}, \sin\frac{\pi}{2} = \frac{1}{8} \times 1 = \frac{1}{8}$$

Sol. 17. (a)

Since the given function is continuous at x = π/2

$$\therefore f(\pi/2) = \lim_{x\to\pi/2} \frac{1-\sin x}{(\pi-2x)^2} \Rightarrow \lambda = \frac{1}{8}$$

Sol. 18. (b)

L.H.L. at x = 3 is 4

R.H.L. at x = 3 is 4

so function is continuous at x = 3

Sol. 19. (d)

L.H.D. at x = 0 is 0

R.H.D. at x = 0 is not 0

so function is not differentiable at x= 0

Sol. 20. (d)

$$f'(x) = \frac{1}{2\sqrt{x+13}} \quad \text{for } x > 3$$

$$f'(12) = \frac{1}{2\sqrt{12+13}} = \frac{1}{10}$$

Sol. 21. (b)

$$\text{We have, } f(x) = \begin{cases} \dfrac{\tan kx}{x} & x < 0 \\ 3x+2k^2 & x \ge 0 \end{cases}$$

Since, f(x) is continuous at x=0

$$\therefore \lim_{x\to 0^-} f(x) = \lim_{x\to 0^+} f(x) = f'(0)$$

$$\Rightarrow \lim_{x\to 0^-}\left(\frac{\tan kx}{x}\right) = \lim_{x\to 0^+}(3x+2k^2) = 3(0)+2k^2$$

$$\Rightarrow \lim_{x\to 0^-}\left[\frac{\tan k(0-h)}{(0-h)}\right] = \lim_{x\to 0^+}\left[3(0+h)+2k^2\right] = 2k^2$$

$$\Rightarrow \lim_{x\to 0^-}\left(\frac{\tan kh}{kh}\right) = 2k^2$$

$$\Rightarrow \lim_{x\to 0^-}\left(\frac{k\tan kh}{kh}\right) = 2k^2 \Rightarrow k = 2k^2\left[\because \lim_{x\to 0}\frac{\tan x}{x} = 1\right]$$

$\Rightarrow$k=1/2 [$\because$k≠0]

Sol. 22. (b)

1. we know that, the greatest integer function is continuous at all points except integer. Therefore, Statement 1 is false.

2. Let h (x) = sin x and g (x) = |x|

$\Rightarrow$ hog (x) = sin |x|

Now, let f (x) = hog (x) = sin |x|

Since, g (x) is continuous, ∀x ∈ R and h (x) is continuous, ∀x ∈R.

$\therefore$ hog (x) is continuous

[$\because$ composition of two continuous functions is continuous]

Hence, f(x) is continuous, ∀x ∈R.

Sol. 23. (a)

$$\text{We have } f(x) = \begin{cases} \dfrac{\alpha\cos x}{\pi-2x} & \text{if } x \ne \dfrac{\pi}{2} \\ 3, & \text{if } x = \dfrac{\pi}{2} \end{cases}$$

Which is continuous at $x = \dfrac{\pi}{2}$

$$\Rightarrow \lim_{x\to\pi/2} f(x) = f\left(\frac{\pi}{2}\right) = 3 \quad \text{....(i)}$$

Now, consider $\lim_{x\to\pi/2} f(x) = \lim_{x\to\pi/2} \dfrac{\alpha\cos x}{\pi-2x}$

$$=\alpha \lim_{x\to\pi/2} \frac{\cos x}{\pi-2x}$$

Now, put y = x – π/2, then as x → π/2 ⇒ y → 0

$$\therefore \lim_{x\to\pi/2} f(x) = \alpha \lim_{y\to\pi/2} \frac{\cos\left(f+\dfrac{\pi}{2}\right)}{\pi-2\left(\dfrac{\pi}{2}+y\right)}$$

$$=\alpha \lim_{y\to 0} \frac{-\sin y}{-2y} = \frac{\alpha}{2}\lim_{y\to 0}\frac{\sin y}{y} = \frac{\alpha}{2} \quad \text{....(ii)}$$

From eqs. (i) and (ii), we get α/2 =3 ⇒ α=6

Sol. 24. (d)

$$\text{Consider } \lim_{x\to 0} f(x) = \lim_{x\to 0} \frac{6\cos x}{\pi-2x} = 6\lim_{x\to 0}\frac{\cos x}{\pi-2x}$$

$$= 6\left(\frac{\cos 0°}{\pi-2(0)}\right) = \frac{6}{\pi} \quad [\because \quad 1]$$

Sol. 25. (a)

L.H.L. at x = 0 is −1

R.H.L. at x = 0 is b

given that function is continuous at x = 0

so b = −1

L.H.L. at x = 1 is 1

R.H.L. at x = 1 is a + b

given that function is continuous at x = 1

so a + b = 1

then a = 2

Sol. 26. (d)

By above solution

Sol. 27. (c)

Both statements are correct

Sol. 28. (a)

At x = –1

LHL = $\lim_{h\to 0} f(-1-h) = \lim_{h\to 0} = a(-1-h)-2$

= – 2 = –a –2

RHL = $\lim_{h\to 0} f(-1+h) = \lim_{h\to 0} (-1) = -1$

At x=1, LHL = $\lim_{h\to 0} f(1-h) = \lim_{h\to 0} (-1) = -1$

RHL= $\lim_{h\to 0} f(1+h) = \lim_{h\to 0} (a+2h^2) = 0$

Now for f to be continuous at x = –1, we should have LHL = RHL = f(–1) ⇒ – a – 2 = –1

⇒ a = – 1

Also for a = –1, f is continuous at x = 1. Hence a = –1

Sol. 29. (c)

Clerly, for f to be continuous at x = π,

$$f(\pi) = \lim_{x\to\pi} \frac{1-\sin x+\cos x}{1+\sin x+\cos x} \quad \left[\frac{0}{0}\text{form}\right]$$

$$= \lim_{x\to\pi} \frac{-\cos x-\sin x}{\cos x+\sin x} \quad [\text{by L'Hospita's rule}]$$

$$= \frac{-\cos\pi-\sin\pi}{\cos\pi-\sin\pi} = \frac{-(-1)-0}{-1-0} = -1$$

Sol. 30. (b)

$$\text{1.consider, } \lim_{h\to 0}\frac{f(0+h)-f(0)}{h} \lim_{h\to 0}\frac{1/h-0}{h}$$

$$= \lim_{h\to 0}\frac{1}{h^2} = \infty \text{(not defined)}$$

Hence, 1 is not differentiable at x = 0

$$\text{2.We have, } f(x) = \begin{cases} 2x+5 & x > 0 \\ x^2+2x+5 & x \le 0 \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} 2 & x > 0 \\ 2x+2 & x \le 0 \end{cases}$$

$\because$ LHL of f' (x) = RHL of f' (x)

$\therefore$ f is differentiable at x = 0

Sol. 31. (c)

Case I For continuity at $x = \dfrac{\pi}{2}$

A + B = 0 ……(i)

Case II For continuity at x = $-\dfrac{\pi}{2}$

2 = - A + B …….(ii)

By both equation A = - 1 and B =1

Sol. 32. (a)

Check above solution

Sol. 33. (c)

LHL= $\lim\limits_{h\to 0^-} f(x) = \lim\limits_{h\to 0} (x+\pi) = \pi$

RHL= $\lim\limits_{h\to 0^+} f(x) = \lim\limits_{h\to 0} \pi\cos x$

$= \pi\cos(0)=\pi$

Also, $f(0) = \pi \cos (0) = \pi$

Hence, f(x) is continuous at x = 0

Statement 1 is correct.

Now, for $x=\pi/2$

LHL = $\lim\limits_{x\to\pi/2^-} f(x) = \lim\limits_{x\to\pi/2} \pi\cos x = \pi\cos\frac{\pi}{2} = 0$

RHL= $\lim\limits_{x\to\pi/2^+} f(x) = \lim\limits_{x\to\frac{\pi}{2}} \left(x-\frac{\pi}{2}\right)^2 = \left(\frac{\pi}{2}-\frac{\pi}{2}\right)^2 = 0$

Also, $f\left(\frac{\pi}{2}\right) = \pi\cos\frac{\pi}{2} = 0$

Hence, f(x) is continuous at $x=\pi/2$Statement 2 is correct

Sol. 34. (d)

LHD $\neq$ RHD

f (x) is not differentiable at x = 0

$\therefore$ statement 1 is not correct.

LHD $\neq$ RHD

f (x) is not differentiable at $x = \frac{\pi}{2}$.

Statement 2 is not correct.

Sol. 35. (b)

modulus functions are always continuous , not differentiable at those points where modulus contain zero.

Sol. 36. (c)

sinx is always continuous but [x] is discontinuous at all integers.

Sol. 37. (d)

(fog) (x) = f(g(x)) = f (sin x) = [sin x]

LHL= $\lim\limits_{x\to 0^-} [\sin x] = \lim\limits_{x\to 0^-} [\sin(0-h)] = -2$

RHL= $\lim\limits_{x\to 0^+} [\sin x] = \lim\limits_{x\to 0^+} [\sin(0+h)] = 0$

$\therefore \lim\limits_{x\to 0}$ (fog) (x) does not exist

and $\lim\limits_{x\to 0}$ (gof) (x)

$= \lim\limits_{x\to 0} g(f(x)) = \lim\limits_{x\to 0} g([x]) = \lim\limits_{x\to 0} \sin[x]$

LHL $\lim\limits_{x\to 0^-} \sin[x] = \lim\limits_{x\to 0^-} [0-h] = \lim\limits_{x\to 0^-}$ $\sin[-h] = -\sin 1$

RHL $\lim\limits_{x\to 0^+} \sin[x] = \lim\limits_{x\to 0^+} \sin[0+h] = \sin(0) = 0$

$\therefore \lim\limits_{x\to 0}$ (gof) (x) does not exist

Clearly, we have

$\lim\limits_{x\to 0^+} [\sin x] = \lim\limits_{x\to 0^+} \sin[x] = 0$

$\therefore \lim\limits_{x\to 0}$ (fog) (x) = $\lim\limits_{x\to 0^+}$ (gof) (x)

Sol. 38. (d)

I. (fof) (x) = f(f(x)) = f([x]) = ([x]) = [x] = f(x)

Which is correct

II. (gog) (x) = g (g(x)) = g (sin x) = sin (sin x)

When x = 0, g(x) = sin x = 0 and (gog) (x)= sin(sin x) = 0,

Which is correct

III. (go(fog)) (x) = g ([sin x]) = sin ([sin x]),

Which can take only three values.

Sol. 39. (b)

Given $f(x) = \begin{cases} 2+x & x \geq 0 \\ 2-x & x < 0 \end{cases}$

At x = 1, LHL =

$\lim\limits_{x\to 1^-} (2+x) = \lim\limits_{h\to 0} (2+(1-h)) = 3$

RHL = $\lim\limits_{x\to 1^+ (2+x)} = \lim\limits_{h\to 0} (2+(1+h)) = 3$

Here LHL = RHL

$\therefore \lim\limits_{x\to 1} f(x)$ exists.

At x = 0, graph of (x) makes sharp edge

$\therefore$ f(x) is not differentiable at x = 0.

At x = 0

LHL = $\lim\limits_{x\to 0^-} (2-x) = \lim\limits_{h\to 0} [2-(0-h)] = 2$

RHL = $\lim\limits_{x\to 0^+} (2+x) = \lim\limits_{h\to 0^-} (2+(0+h)) = 2$

Values of f(x) at x = 0

i.e. f(0) = 2 + 0 = 2

Here, LHL = RHL = f(x)

$\therefore$ f(x) is continuous at x = 0

Sol. 40. (d)

We have

$F(x) = \begin{cases} -2, & -3 \leq x \leq 0 \\ x-2 & 0 < x \leq 3 \end{cases}$

$\Rightarrow(|x|) = \begin{cases} -2, & -3 \leq |x| \leq 0 \\ |x|-2 & 0 < |x| \leq 3 \end{cases}$

At $-3 \leq |x| \leq 0$, f (|x|) = −2 not possible

$\Rightarrow(|x|) = \begin{cases} -x-2, & -3 \leq x \leq 0 \\ x-2 & 0 < x \leq 3 \end{cases}$

and $|f(x)| = \begin{cases} |-2| & -3 \leq x \leq 0 \\ |x-2| & 0 < x \leq 3 \end{cases}$

$\Rightarrow |f(x)| = \begin{cases} 2 & -3 \leq x \leq 0 \\ (2-x) & 0 < x < 2 \\ (x-2) & 2 \leq x \leq 3 \end{cases}$

From eqs. (i) and (ii) we get

g(x) = f(|x|) + |f(x)|

$g(x) = \begin{cases} -x & -3 \leq x \leq 0 \\ 0 & 0 < x < 2 \\ 2x-4 & 2 \leq x \leq 3 \end{cases}$ $g'(x) = \begin{cases} -1 & -3 \leq x \leq 0 \\ 0 & 0 < x < 2 \\ 2 & 2 \leq x \leq 3 \end{cases}$

Since, g (x) has sharp edges at x = 0 and x = 2

Hence, g(x) is not differentiable at x = 0 and 2

Sol. 41. (a)

Check above solution

g' (x) = − 1

Sol. 42. (d)

Check above solution

Sol. 43. (a)

We have f:A$\to$R, where A=R/{0}

such that, f(x) $\frac{x+|x|}{x}$

$\Rightarrow f(x) = \begin{cases} 2 & x > 0 \\ 0, & x < 0 \end{cases}$

Since, f(x) is not defined for x = 0

$\therefore$ f (x) is discontinuous at x = 0

Also, f (x) is constant function on R − {0}

Hence, f (x) is continuous on A, where A = R − {0}

Sol. 44. (a)

for negative powers of x function is not continuous at x = 0

so function is continuous and differentiable for all positive value of x.

Sol. 45. (c)

L.H.L. at x = 0 is −1

R.H.L. at x = 0 is +2

function is not continuous at x = 0

L.H.L. at x = −2 is −3

R.H.L. at x = −2 is −3

function is continuous at x = −2

Sol. 46. (c)

$f(x) = -\frac{x}{\sqrt{x^2}} = -\frac{x}{|x|}$ when $x \neq 0$

R.H.L. of this function is −1

and L.H.L. = 1

so function is discontinuous at x = 0

Sol. 47. (c)

left hand limit of $\sqrt{x}$ is not defined at x = 0

so function is not continuous at x = 0.

Sol. 48. (a)

1. continuous + continuous
 = continuous
statement is correct.

2.continuous + discontinuous
= discontinuous
statement is correct.

3.continuous + discontinuous
= discontinuous statement is incorrect.

Sol. 49. (a)

L.H.D. at k

$= \lim\limits_{h\to 0} \frac{f(k-h)-f(k)}{-h}$

$= \lim\limits_{h\to 0} \frac{[k-h]\sin(\pi(k-h)) - [k]\sin(k\pi)}{-h}$

here k is integer so $\sin k\pi = 0$

$= \lim\limits_{h\to 0} \frac{[k-h]\sin(\pi(k-h))}{-h}$

$= \lim\limits_{h\to 0} (k-1)\pi \frac{\sin(\pi(k-h))}{-\pi h}$

$= \lim\limits_{h\to 0} (-1)^k (k-1)\pi \frac{\sin(-\pi h)}{-\pi h}$

$= (-1)^k (k-1)\pi$

Sol. 50. (c)

$f(x) = \frac{x}{2} - 1$

$\tan[f(x)] = \tan\left[\frac{x}{2} - 1\right]$

we have to check from x = 0 to x = π

we know that G.I.F. are not continuous at any integer point.

check continuity at x = 2

R.H.L at x = 2 is tan0 = 0

L.H.L. at x = 2 is tan(-1) = - tan1

so function tan[f(x)] is not continuous at x= 2

$f(x) = \frac{x}{2} - 1$

$\frac{1}{f(x)} = \frac{2}{x-2}$

above function is not continuous at x = 2

Sol. 51. (c)

$f(x) = \sqrt{1-e^{-x^2}}$ and $f'(x) = \frac{2xe^{-x^2}}{2\sqrt{1-e^{-x^2}}}$

at x = 0, f '(0) is not defined so function is differentiable except x = 0

Sol. 52. (d)

$f(x) = [x]^2 - [x^2]$

R.H.L. at x =0

0 − 0 = 0

L.H.L. at x = 0

1 − 0 = 1

Function is discontinuous at x = 0

R.H.L. at x =1

OFFICIUM SUPREMUM

$1 - 1 = 0$
L.H.L. at x = 0
$0 - 0 = 0$
Function is continuous at x = 1

Sol. 53. (b)
$\lim_{x\to 3}\frac{(x-3)(x+3)}{(x-3)(x+1)} = \frac{3}{2} = 1.5$

Sol. 54. (a)
sum of two continuous functions is always continuous

Sol. 55. (b)
$\lim_{x\to 0}\frac{2x-\sin^{-1}x}{2x+\tan^{-1}x}$
apply L. hospital
$\lim_{x\to 0}\frac{2-\frac{1}{\sqrt{1-x^2}}}{2+\frac{1}{1+x^2}} = \frac{1}{3}$

Sol. 56. (a)
$|x-3|$ is always continuous.

Sol. 57. (a)
$\lim_{x\to 0}\frac{\sin 2x}{5x}$
apply L hospital rule
$\lim_{x\to 0}\frac{2\cos 2x}{5} = \frac{2}{5}$
but given that f(0) = 2/15
so function is not continuous at x = 0

Sol. 58. (d)
f(0 + h) = f(0)
$\lim_{h\to 0}\sin(0+h) = k = 0$

Sol. 59. (a)
f(0 + h) = f(0)
$\lim_{h\to 0}\left(0+h+\frac{1}{2}\right)^2 = k$
$\left(\frac{1}{2}\right)^2 = k = \frac{1}{4}$

Sol. 60. (d)
$\lim_{x\to 0}\sin\left(\frac{1}{x}\right)$ not exists because it oscillate between −1 and 1.

Sol. 61. (a)
$f(x) = e^{-|x|}$
|x| is continuous at x = 0 but not differentiable at x = 0

Sol. 62. (b)
Standard limit

Sol. 63. (a)
$f(x) = \begin{cases} a+bx, & x<1 \\ 5, & x=1 \\ b-ax, & x>1 \end{cases}$
function is continuous
so, L.H.L. = f(1) = R.H.L.
Left hand limit
f(1− 0) = a + b
f(1) = 5
so a + b = 5

Sol. 64. (a)
F(x) = |x|–1
1.f(x) is continuous at x=1
2.f(x) is differentiable at x=0
f(x)= $\begin{cases} -x-1; & x\le 0 \\ x-1; & x>0 \end{cases}$
so for continuity $\lim_{x\to a^-}$ = f(x)=f(a) = $\lim_{x\to a^+}$ f(x)
here $\lim_{x\to 1}$ x–1 = 1–1 = 0
f(x) = x – 1
f(i) = 1–1 = 0
$\lim_{x\to 1^-}$ f(x) = (1) = $\lim_{x\to 1^+}$ f(x)
So function is breaking at 0 so it has no problem of continuity on any other point on real line so function is continuous at x = 1 for differentiability Lf' (a) = Rf' (a)
Here a = 0
Lf'(1) ie $\lim_{x\to 1^-}$
Lf'(0) = – 1
Rf'(0) = 1
Lf' (0) ≠ Rf'(0)
So function is not differentiable

Sol. 65. (b)
f(x)=$\sin\left(\frac{1}{x^2}\right)$ x≠0
1.I its continuous at x = 0, if f(0) = 0
2.It is continuous at x= $\frac{2}{\sqrt{1}}$
For 1: for checking continuity at x = 0
$\lim_{x\to 0^-}$ f(x) = f(0) = $\lim_{x\to 0^+}$ f(x)
Should be equal
$\lim_{x\to 0}$ f(x)
$\lim_{x\to 0}\sin\left(\frac{1}{x^2}\right)$ =sin ∞
f(0) = 0 so they are not equal
(1)statement is incorrect
$\lim_{x\to \frac{2}{\sqrt{\pi}}} f\left(\frac{2}{\sqrt{\pi}}\right) = \frac{1}{\sqrt{2}}$
$\sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$ so $\lim_{x\to \frac{2}{\sqrt{\pi}}} = f\left(\frac{2}{\sqrt{\pi}}\right)$
so f(x) is continuous at x= $\frac{2}{\sqrt{\pi}}$

Sol. 66. (a)
y = [x]
this function is discontinuous and not differentiable and not differentiable only on integer points. And on fractional points this function is constant function and its derivative will be zero.

Sol. 67. (b)
f(x)= $\begin{cases} x+6, x\le 1 \\ px+q, 1<x<2 \\ 5x, x\ge 2 \end{cases}$
continuous at x = 1
1 + 6 = p + q …..(i)
Continuous at x = 2
2p + q = 10
By both equations p = 3 , q = 4

Sol. 68. (c)
Check above solution

Sol. 69. (d)
f(x) = $\begin{cases} \frac{x-3}{|x-3|}+a; & x<3 \\ a-b; & x=3 \text{ and} \\ \frac{x-3}{|x-3|}+b; & x>3 \end{cases}$
L.H.L at x = 3 is - 1+ a
R.H.L. at x = 3 is 1 + b
f(3) = a – b
f(x) is continuous so
-1 + a = 1 + b = a – b
a = 1 and a = 3

Sol. 70. (b)
Check above solution

Sol. 71. (c)
f(x) = $\begin{cases} ax(x+1)+b & x<1 \\ x-1, & 1\le x\le 2 \end{cases}$
f '(x) = $\begin{cases} 2ax+a & x<1 \\ 1 & 1\le x\le 2 \end{cases}$
function is differentiable at x = 1
L.H.D. at x= 1 is equal to R.H.D. at x = 1
3a + 1 = 1
a = 0
function is continuous at x = 1
L.H.L. at x= 1 is equal to R.H.L. at x = 1
2a + b = 0
If a = 0 then b=0
Then a + b = 0

Sol. 72. (c)
$\lim_{x\to 0}$ ax(x+1)+b = 0+ b = 0

Sol. 73. (a)
f(x) = |x| + 1 , |x| is not differentiable at x = 0 , so f(x) is always differentiable for x < 0
statement 1 is correct.
g(x) = [x] – 1,
[x] is only discontinuous at integer. so g(x) is continuous at x = 0.0001.
[x] is not differentiable at integer points, at any other point this function behaves like a constant function.
so derivative of g(x) = 0
statement 3 is incorrect.

Sol. 74. (b)
$f(x) = [x]^2 + [x^2]$
at x= 0
Right hand limit
$= \lim_{h\to 0}$ $[0+h]^2 - [(0+h)^2] = [h]^2 - [h^2]$
= 0 – 0 = 0
left hand limit
$= \lim_{h\to 0}$ $[0-h]^2 - [(0-h)^2] = [-h]^2 - [h^2]$
$= (-1)^2 - 0 = 1$
f(x) is not continuous at x = 0
at x= 1
Right hand limit
$= \lim_{h\to 0}$ $[1+h]^2 - [(1+h)^2] = [1+h]^2 - [(1+h)^2]$
= 1 – 1 = 0
left hand limit
$= \lim_{h\to 0}$ $[1-h]^2 - [(1-h)^2] = [1-h]^2 - [h^2]$
= 0 – 0 = 0
f(x) is continuous at x = 1

Sol. 75. (c)
$f(x) = \begin{cases} x^3 & x^2<1 \\ x^2, & x^2\ge 1 \end{cases} \Rightarrow f'(x) = \begin{cases} 3x^2 & x^2<1 \\ 2x, & x^2\ge 1 \end{cases}$
f'(0) = 3(0) = 0

Sol. 76. (d)
$f(x) = \begin{cases} x^2 & x\le -1 \\ x^3 & -1<x<1 \\ x^2 & x\ge 1 \end{cases}$
Function is continuous at x = 1 but not continuous at x = − 1
$f'(x) = \begin{cases} 3x^2 & -1<x<1 \\ 2x & x\ge 1 \end{cases}$
Function is not differentiable at x = 1
Function is not differentiable at x = 1

Sol. 77. (a)
Function is continuous
So $\lim_{x\to 0^-} f(x) = \lim_{x\to 0^+} f(x)$
$\lim_{x\to 0^-} 4\left(5^x\right) = \lim_{x\to 0^+}\left[8k + x\right]$
$\lim_{h\to 0} 4\left(5^{-h}\right) = \lim_{h\to 0}\left[8k + (0+h)\right]$
4 = 8k
k = 0.5

Sol. 78. (d)
At x = 3
for 3 < x < 4, f(x) = x – 3 – x + 4 = 1
for x < 3 , f(x) = – x + 3 – x + 4 = –2x + 7
right hand derivative at 3 = 0
left hand derivative at 3 = – 2
so function is not differentiable at x = 3
now at x = 4
for 3 < x < 4, f(x) = x – 3 – x + 4 = 1
for x > 4 , f(x) = x – 3 + x – 4 = 2x – 7
left hand derivative at 4 = 0
right hand derivative at 4 = 2
so function is not differentiable at x = 4